Agreement Meets Confidence: A Training-Free Fusion That Outperforms EigenScore for Hallucination Detection

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***Abstract*—Large language models often state wrong facts with full confidence, and detecting such hallucinations without a reference answer remains difficult. EigenScore, from the INSIDE framework, samples several answers to a question and measures how widely their hidden states spread. We first reproduce EigenScore on an 8 GB consumer GPU. We then show that the same samples carry two further, partly independent kinds of evidence, namely how much the answers agree in their wording and how confident the model is in each token, and we propose a label-free rank fusion of the three signals. Following a development and held-out protocol in which all methods were frozen before the test data were analysed, the fusion outperforms EigenScore by 2.7 points of AUROC on both held-out datasets and by 4.5 points on the development set, with Llama-2-7B on 1,000 questions each from CoQA, TriviaQA and Natural Questions. It is the only method that significantly improves on EigenScore on every dataset, at no additional generation cost.**

***Index Terms*—Hallucination detection, large language models, rank fusion, uncertainty estimation, internal states.**

I. Introduction

LARGE language models (LLMs) state wrong facts in the same fluent tone as correct ones, so a reader cannot separate the two. A hallucination detector gives each answer a score without access to the correct answer. Existing detectors read one of three kinds of evidence:

- **Token confidence** of the generated words, e.g. perplexity or length-normalised entropy [5].
- **Surface agreement** between several sampled answers, measured on their text [4] or meaning [3].
- **Internal states** of the model [1], [2].

INSIDE [1] reads internal states. Its *EigenScore* samples K answers and scores the log-volume of their hidden-state embeddings. It therefore sees only one kind of evidence: when the model repeats the same wrong answer K times, the spread collapses and EigenScore trusts it, even though the model's token confidence may still be low.

Contributions.

1) A working reproduction of INSIDE, whose public code did not run, on an 8 GB laptop GPU.
2) **TRI**, a training-free rank fusion of internal, surface and token evidence computed from the same samples, together with **CTC**, a tie-breaking variant with a closed-form gain.
3) A development/held-out evaluation in which every method was committed to version control before the held-out data were analysed. TRI is the only method that significantly beats EigenScore on all three datasets.

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II. Related Work

Token-level uncertainty. Perplexity and length-normalised entropy [5] score a single answer from its token probabilities. They are cheap, but a model can be confidently wrong.

Sampling consistency. SelfCheckGPT [4] compares sampled answers on their text, and semantic entropy [3] clusters them by meaning before computing an entropy. Both use agreement between samples, but they ignore the model's token confidence and internal states.

Internal states. INSIDE [1] computes a log-determinant over the embeddings of the sampled answers. LLM-Check [2] applies similar eigen-analysis to the hidden states and attention maps of a single response, and reports that its quality varies across layers. Trained probes on hidden states are accurate but need labelled data [6].

Evaluation. Janiak et al. [7] show that ROUGE-based labels inflate detector AUROC and that simple baselines are often competitive. This motivates our comparison against strong text baselines and our held-out protocol.

Our work differs from these in combining the three kinds of evidence, without training, from the samples that a sampling-based detector already draws.

III. Existing Work: EigenScore

For a question, the model samples K answers. Let $\mathbf{z}_k \in \mathbb{R}^d$ be the hidden state of answer k's last token at the middle layer, $Z = [\mathbf{z}_1, \ldots, \mathbf{z}_K]$, and J_d the feature-centring matrix. With $\Sigma = Z^\top J_d Z/(d-1) \in \mathbb{R}^{K\times K}$,

$$\mathrm{ES} = \frac{1}{K}\sum_{i=1}^{K} \log_{10} \lambda_i(\Sigma + \alpha I)\,, \qquad \alpha = 10^{-3}. \tag{1}$$

For a Gaussian, $\frac{1}{2}\log\det\Sigma$ is the differential entropy up to a constant, so ES estimates the semantic uncertainty of the answers [1].

Reproduction. The released code had twelve blocking defects and lacked the feature-clipping component. After repairing it, we reproduced the paper's method ranking on 250 CoQA questions with Llama-2-7B in 4-bit precision: EigenScore reached 83.5 AUROC, above Lexical Similarity (80.5), LN-Entropy (77.5), Perplexity (72.3) and Energy (54.7).

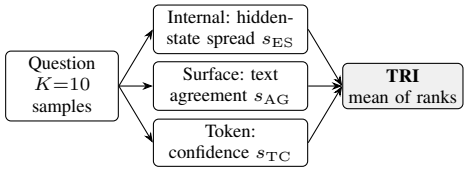


Fig. 1. TRI reads three kinds of evidence from the same K samples and fuses their ranks. No extra generation and no training are needed.

IV. Proposed Method

A. Three signals from the same samples

For each question we draw $K = 10$ samples and also decode a greedy answer. We compute three scores, each oriented so that a higher value means a more likely hallucination:

- **Internal** – s_{ES}: EigenScore at layer 16.
- **Surface** – s_{AG}: minus the mean pairwise ROUGE-L between the K samples (text agreement).
- **Token** – s_{TC}: LN-Entropy, the mean over samples of the per-token negative log-likelihood.

None requires extra generations: the hidden states and log-probabilities are recorded during sampling.

B. TRI: rank fusion

The three scores live on unrelated scales. We therefore fuse their ranks $r(\cdot)$ over the questions:

$$\mathrm{TRI} = \tfrac{1}{3}\left[r(s_{\mathrm{ES}}) + r(s_{\mathrm{AG}}) + r(s_{\mathrm{TC}})\right]. \tag{2}$$

The weights are equal and nothing is fitted, so TRI needs no labels.

C. CTC: consistency, then confidence

Text agreement is exactly tied on many questions, mostly where all K samples are identical. AUROC counts a tied correct/incorrect pair as one half: $\mathrm{AUROC} = P(s_1 > s_0) + \frac{1}{2}P(s_1 = s_0)$. CTC orders questions by s_{AG} and breaks exact ties by s_{TC}. Every untied pair keeps its order, so the change in AUROC is exactly

$$\Delta\mathrm{AUROC} = (a - \tfrac{1}{2})\, P(\text{tied pair}), \tag{3}$$

where a is the AUROC of s_{TC} within the tied pairs. CTC thus targets the self-consistent case that EigenScore misses: when every sample agrees, token confidence is the only evidence left.

V. Experimental Protocol

- **Model.** Llama-2-7B in 4-bit NF4 on an RTX 4060 laptop GPU (8 GB). $K = 10$ samples per question at temperature 0.5, top-p 0.99 and top-k 5 [1].
- **Data.** 1,000 random questions each from CoQA (story and previous turns in the prompt), TriviaQA (no context) and NQ-Open.
- **Labels.** An answer is a hallucination when its greedy answer has ROUGE-L ≤ 0.5 against every reference.
- **Development and held-out split.** All methods were developed on CoQA only. TRI and CTC were then committed to git together with their hypotheses, before any TriviaQA or NQ-Open result was computed.
- **Statistics.** AUROC per method; paired bootstrap over questions (10,000 resamples); Holm correction over all held-out superiority tests. A claim counts as confirmed only if it holds on both held-out datasets.

TABLE I
AUROC (%) ON 1,000 QUESTIONS PER DATASET. COQA WAS THE DEVELOPMENT SET; TRIVIAQA AND NQ-OPEN WERE HELD OUT.

Method	CoQA	TriviaQA	NQ-Open	Mean
Perplexity	74.6	79.8	74.8	76.4
LN-Entropy (token)	77.8	80.6	76.0	78.1
EigenScore (internal) [1]	80.1	80.5	76.7	79.1
Text agreement (surface)	85.3	82.5	76.6	81.5
CTC (ours)	**85.7**	**83.7**	77.1	82.2
TRI (ours)	84.6	83.2	**79.5**	**82.4**

Model accuracy: CoQA 81.4%, TriviaQA 69.9%, NQ-Open 26.9%.

TABLE II
HELD-OUT CONFIRMATION: ΔAUROC (POINTS) WITH 95% BOOTSTRAP CI. ALL TESTS WERE FIXED BEFORE THE HELD-OUT DATA WERE ANALYSED.

Comparison	TriviaQA	NQ-Open	Verdict
TRI vs EigenScore	+2.7 [+1.7, +3.8]	+2.7 [+1.3, +4.2]	confirmed
TRI vs text agreement	+0.7 [−0.4, +1.8]	+2.9 [+1.3, +4.5]	non-inferior
CTC vs text agreement	+1.2 [+0.6, +1.8]	+0.5 [+0.2, +0.7]	confirmed
CTC vs EigenScore	+3.2 [+1.7, +4.8]	+0.3 [−1.9, +2.6]	partial
Text agreement vs ES	+2.1 [+0.6, +3.5]	−0.1 [−2.4, +2.2]	partial

VI. Results

Main result. Table I shows that TRI has the highest mean AUROC, and Table II that it beats EigenScore by 2.7 points on both held-out datasets, significant after Holm correction. On CoQA the gain is 4.5 points (95% CI $[+2.7, +6.2]$). No single signal is reliable everywhere: text agreement is the strongest signal on CoQA but no better than EigenScore on NQ-Open, where the model answers only 27% of the questions correctly. There every single signal stays below 77 AUROC, while TRI reaches 79.5.

CTC. CTC beats text agreement on both held-out datasets. Its gains match the closed form of Section III-C: within tied pairs, token confidence reaches $a = 0.73$, 0.70 and 0.65 on CoQA, TriviaQA and NQ-Open, on 1.9%, 5.9% and 3.1% of all correct/incorrect pairs. On CoQA, EigenScore reaches only $a = 0.59$ within the same ties, which confirms that internal spread carries little information once the answers agree.

Why fusion helps. The three signals are only partly correlated: Spearman correlations range from 0.62 (agreement vs confidence on NQ-Open) to 0.90 (EigenScore vs agreement on TriviaQA). Internal spread and text agreement overlap most, and token confidence is the most independent. A post-hoc ablation agrees. On NQ-Open, dropping text agreement or token confidence from TRI costs 1.1–1.2 points (both significant), and dropping EigenScore costs 0.6 points (CI $[-0.3, +1.4]$). On CoQA and TriviaQA, agreement plus confidence alone

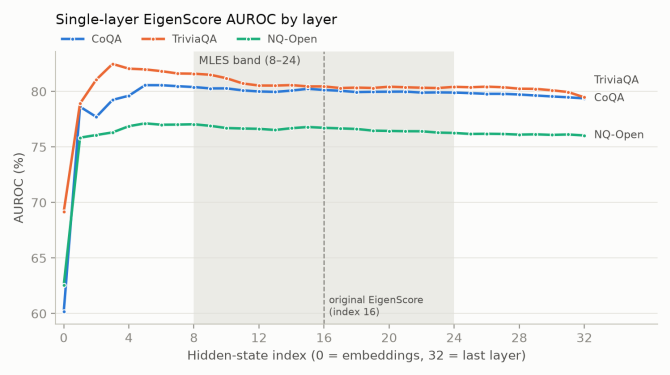


Fig. 2. AUROC of single-layer EigenScore at every depth. From layer 5 onward, performance hardly depends on the layer.

matches TRI. The dominant effect is therefore the combination of agreement-type evidence with token confidence.

Layer choice. Fig. 2 shows single-layer EigenScore at every depth. From layer 5 onward the curve is almost flat, so INSIDE's middle-layer choice is not fragile. Accordingly, averaging EigenScore over layers 8–24 changed AUROC by only -0.1 point on every dataset.

What did not help. Before arriving at the fusion, we also tested several EigenScore variants on all three datasets:

- Multi-layer averaging: -0.1 point on every dataset.
- Normalising the embeddings: between -0.6 and $+0.9$ points, not significant.
- A cross-layer stability score based on representational similarity: -1.8 to -4.3 points.

These results point the same way as the fusion: the gains come from adding a different kind of evidence, not from refining the internal-state signal.

VII. Discussion and Limitations

- **Labels.** Correctness is judged by ROUGE-L, which has low precision and inflates the AUROC of sampling-based detectors [7]. Paired differences share the same labels and are less affected, but an LLM-judge evaluation is needed.
- **Scope.** We study one 7B model in 4-bit precision and 1,000 questions per dataset.
- **Cost.** Like EigenScore, TRI needs K generations per question. The fusion itself is free.
- **Open-weight models only.** The internal signal requires hidden states. Agreement plus confidence, which carries most of the gain, also works with APIs that expose token log-probabilities.

VIII. Conclusion

A label-free rank fusion of internal, surface and token evidence, computed from the samples EigenScore already draws, outperforms EigenScore by 2.7 to 4.5 AUROC points on three question-answering datasets. The gains held on two datasets that were frozen out of development. TRI is the only method that significantly improves on EigenScore on every dataset. The main lesson is that confidence and agreement fail on different questions, so a detector should read both. Future work will test a second model family and LLM-judge labels.

References

[1] C. Chen, K. Liu, Z. Chen, Y. Gu, Y. Wu, M. Tao, Z. Fu, and J. Ye, "INSIDE: LLMs' internal states retain the power of hallucination detection," in *Proc. ICLR*, 2024.

[2] G. Sriramanan, S. Bharti, V. S. Sadasivan, S. Saha, P. Kattakinda, and S. Feizi, "LLM-Check: Investigating detection of hallucinations in large language models," in *Proc. NeurIPS*, 2024.

[3] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, pp. 625–630, 2024.

[4] P. Manakul, A. Liusie, and M. J. F. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," in *Proc. EMNLP*, 2023.

[5] A. Malinin and M. Gales, "Uncertainty estimation in autoregressive structured prediction," in *Proc. ICLR*, 2021.

[6] Z. Zhang *et al.*, "ICR Probe: Tracking hidden state dynamics for reliable hallucination detection in LLMs," in *Proc. ACL*, 2025.

[7] D. Janiak, J. Binkowski, A. Sawczyn, B. Gabrys, R. Shwartz-Ziv, and T. Kajdanowicz, "The illusion of progress: Re-evaluating hallucination detection in LLMs," arXiv:2508.08285, 2025.